Kowalski, analysis!

Introduction to Survival Analysis

Kaplan-Meier Survival Curves & Cox PH Models

Remember to download the data and code, Skipper!

Florida Entomological Society Annual Meeting 2026

St. Augustine

July 15th, 2026

1

Today

- Theory
  - Basic concepts of Survival Analysis
  - Kaplan-Meier (KM) Curves
  - Cox Proportional Hazards (PH) Model
- R Example

2

What is survival analysis?

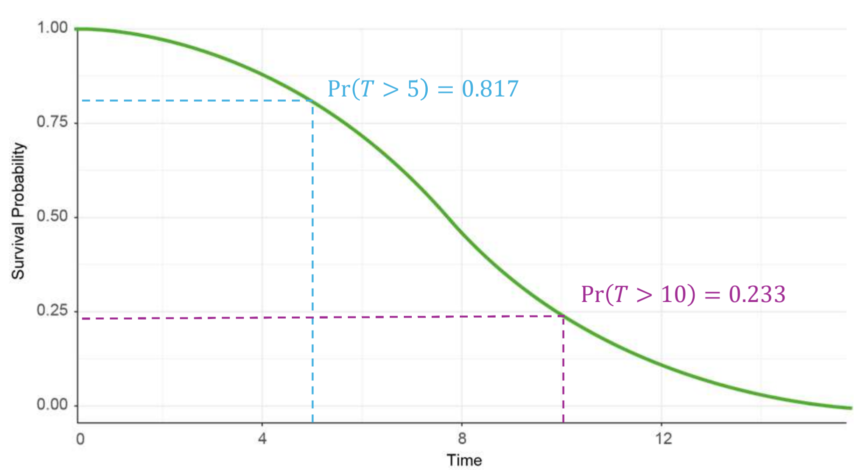
→ Time-to-event analysis: investigating how long it takes for an event to occur

- Outcome often referred to as either **survival time**, **failure time**, or **time to event**
- Examples in entomology:
  - **Insecticide Susceptibility & Toxicology**
    e.g., time until insect death after insecticide exposure
  - **Insect Development & Life Cycle Analysis**
    e.g., time to emergence or pupation
  - **Impact of Biotic and Abiotic Factors**
    e.g., time until death under different temperature regimes

3

Survival function

$$S(t) = \Pr(T > t) = 1 - P(t)$$



Survival function *S(t)*

Expresses the probability that a subject's true survival* time will exceed time , i.e., that the subject will survive* beyond time *t.*

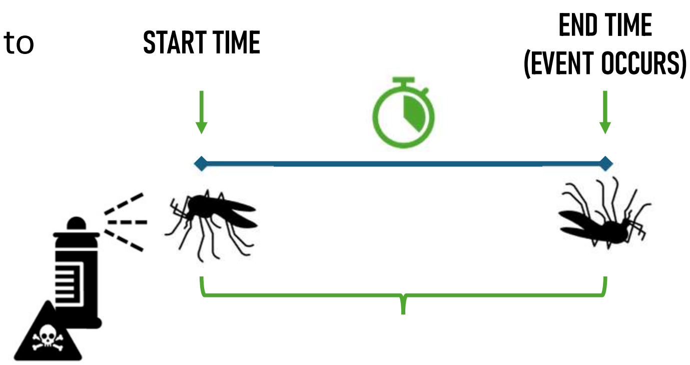
* i.e., probability that the subject will not have experienced the event by time *t*.

4

Example: Survival after insecticide exposure

Information of interest: **duration**

- **Start point**: first exposure to insecticide
- **End point**: death

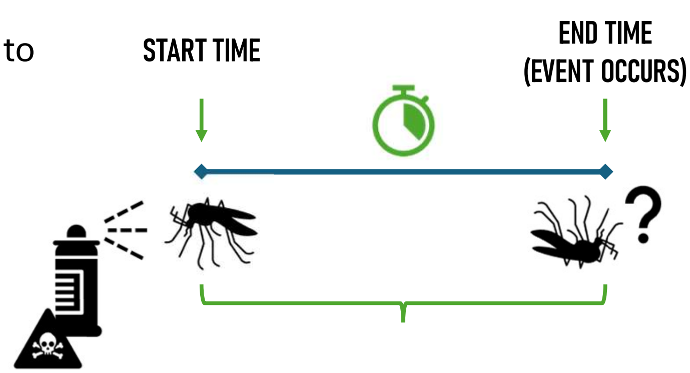


5

Example: Survival after insecticide exposure

Information of interest: **duration**

- **Start point**: first exposure to insecticide
- **End point (realistically):**
  - death OR
  - study ends OR
  - subject is censored



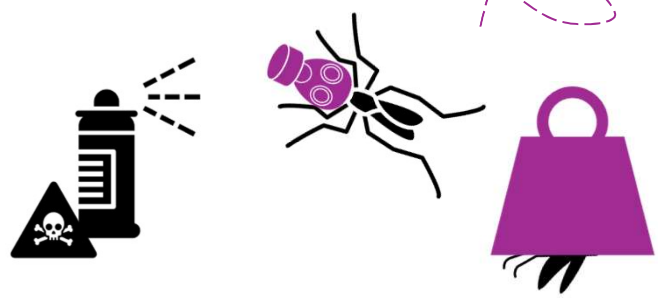
6

What is censoring?

- **Censoring** is a type of missing data problem unique to survival analysis.
- To estimate **survival probability**: Important to **include censored data** in analysis

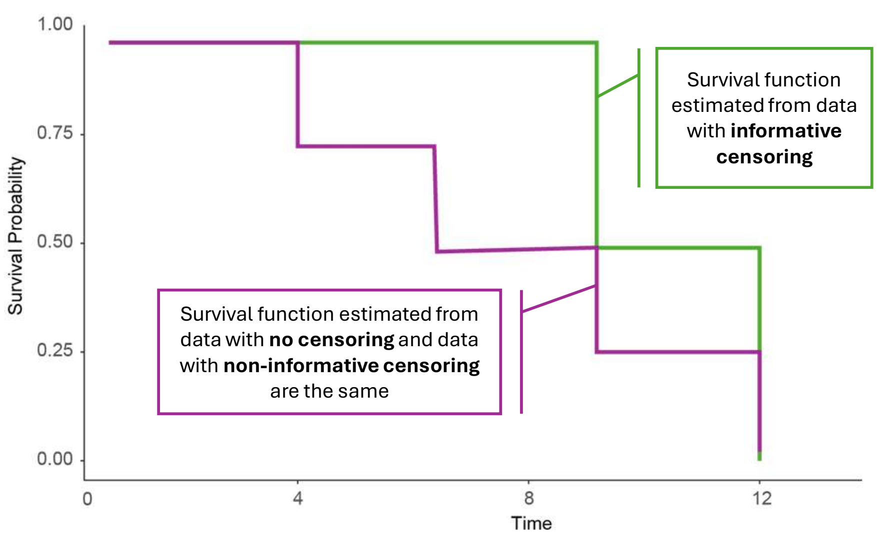
Insecticide example

- Mosquito still alive at study end
- Escaped or lost during handling
- Death due to unrelated reasons



7

Assumption of noninformative censoring



Assumptions

- subject's censoring time not related to the unobserved survival time,
- distribution of censoring times and survival times are unrelated

8

Common methods for survival analysis

	Kaplan-Meier (KM)	Cox Proportional Hazard (PH)
Test	parametric	semi-parametric
Pros	- simple to interpret - can **estimate S(t)**	- can **estimate** hazard ratio **HR** - **hazard** can **fluctuate with time**
Cons	- no functional mathematical form - can **not estimate** a **HR** - only **few** and **categorical** independent variables	- can **not estimate S(t)**

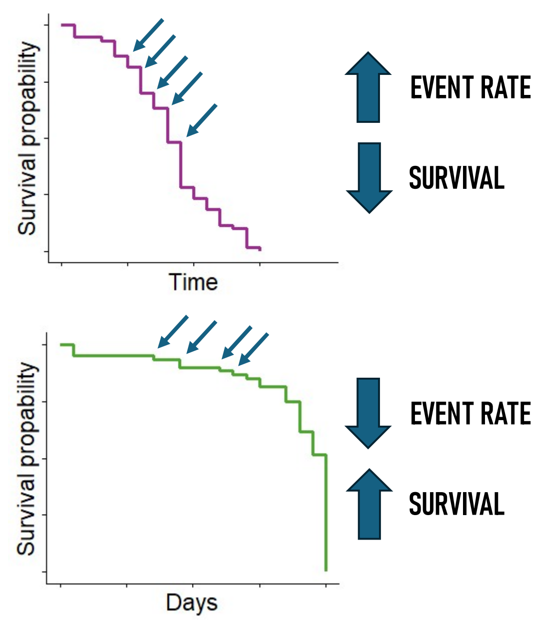
9

Kaplan-Meier Curves

Kaplan-Meier (KM) curves estimate **survival probability** over time

→ **Visual tool** to compare groups

- X-axis: Time
- Y-axis: Probability of survival
- Each step = death event
- Tick marks = censored data

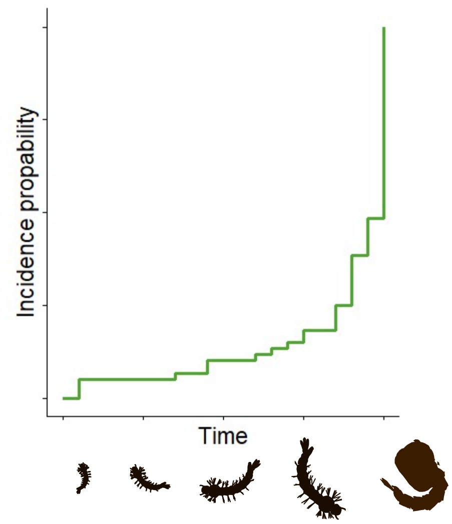


10

Kaplan-Meier Curves

Kaplan-Meier (KM) curves can also plot **cumulative event probability** (or cumulative incidence)

Cumulative events (f(y) = 1-y) **if** event can only be experienced once



11

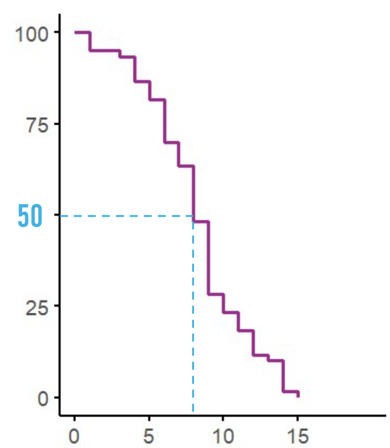
Kaplan-Meier Curves

Interpreting KM Curves

- Steep = higher death rate
- Flat = better survival
- Tick marks = censored observations

Median survival time:

- *t* at which 50 % of population is expected to *not have experienced the event yet*



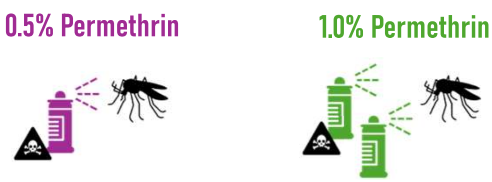
Median survival = 8 time units

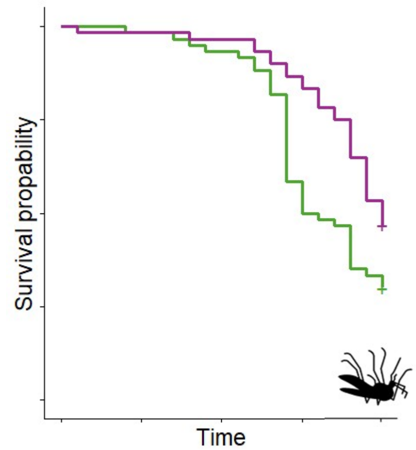
12

Kaplan-Meier Curves

Example: Insecticide exposure

- Compare survival under 0.5% vs. 1.0% permethrin
- Green line drops faster → **higher mortality**





13

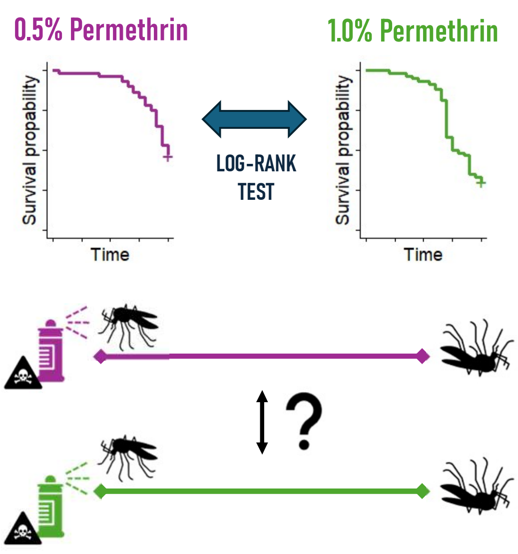
Comparing Groups

- Plot KM curves for:
    - Resistant vs. susceptible strains
    - Field vs. lab mosquitoes
    - Different environmental conditions

But is the difference statistically significant?

→ **Log-rank test**

- **one variable** at a time
- does not estimate **effect size** or **adjust for other factors**



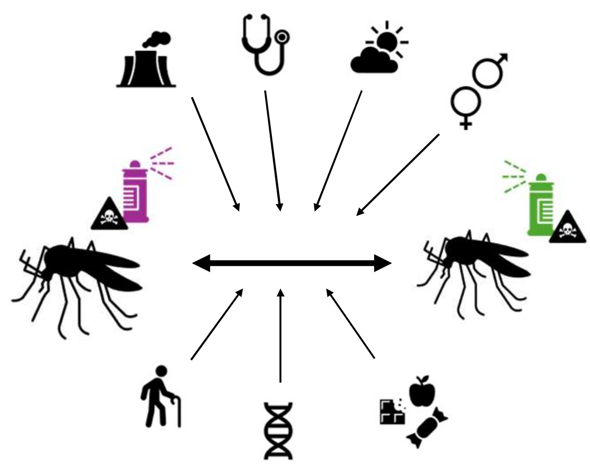
14

Cox Proportional Hazard Model

aka **Cox Regression**

estimates how variables affect the **hazard** – or risk of event – **over time**.

- Allows for censored data
- Multiple predictors possible

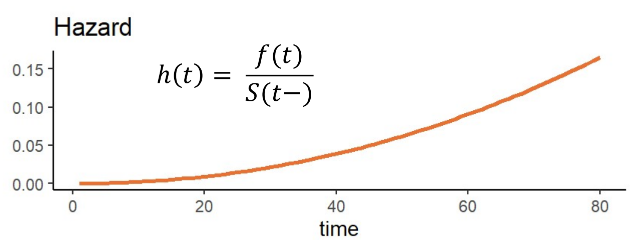


15

Cox Proportional Hazard Model

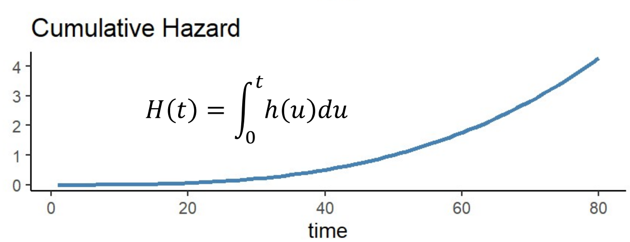
Hazard: probability of the **event** occurring at **a specific time**

$h(\mathrm{t}) \geq 0$, **so hazard can never be negative!**



Cumulative Hazard: hazard a subject has accumulated over time up to time *t*.

$H(\mathrm{t})$ **can never decrease over time!**

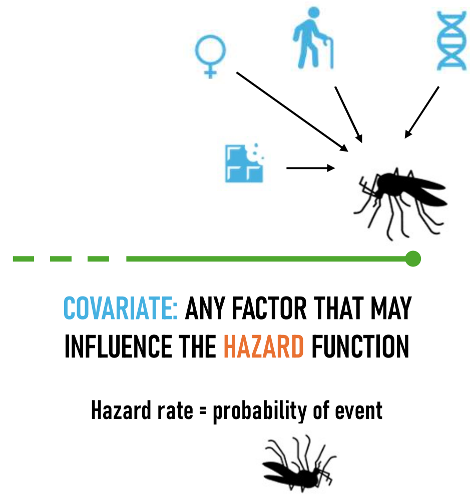


16

Cox Proportional Hazard Model

Multivariate Models

- Include multiple variables:
  - Insecticide dose
  - Mosquito strain
  - Temperature
  - Age or sex
- Adjust for confounding effects



- **Univariate**: one variable at a time (e.g., insecticide dose only)
- **Multivariate**: multiple variables at once (e.g., dose + strain + age)

17

Cox Proportional Hazard Model

expressed by **hazard function:**

$$h(t) = h_0\ (t) \times exp(\beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_k X_k)$$

where:

t ...	survival time
h_0 ...	baseline hazard
$h(t)$ ...	hazard function determined by a set of covariates (X_1, X_2,...)
β ...	effect size of covariates

$$h(t|age) = h_0\ (t) \times exp(\beta_1 X_1 + age * \beta_{age} + \cdots)$$

age at beginning of observation

18

Cox Proportional Hazard Model

Model output: **Coefficients β**

β indicate **log hazard radio,** *assuming all other covariates remain constant*

Example

β = 0.2 → **20% increase** in the **log hazard** of the event per **one-unit increase** of covariate

β > 0: increased log hazard → higher risk

β < 0: decreased log hazard → lower risk

β ~ 0: **no difference** in hazard → variable has no/very little effect on survival time

19

Cox Proportional Hazard Model

Model output: **hazard e^{β}**

- exponentiated form of the coefficient
- **Relative change in hazard** of event for a **one-unit change** in covariate

Hazard ratio: hazard for a group relative to the hazard for the reference group:

$$HR = \frac{H_{x=1}}{H_{x=0}}$$

HR > 1: increased hazard

HR < 1: decreased hazard.

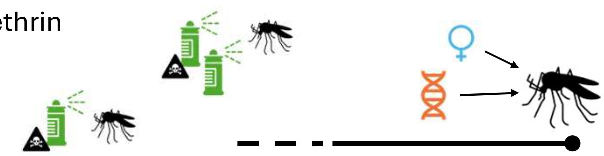
HR = 1: no change in the hazard

20

Cox PH Model – Output Interpretation

COVARIATE	COEFFICIENT	HAZARD RATIO		INTERPRETATION
Insecticide_dose 0.5%, 1.0%	0.588	1.8	>1	⬆ RISK
sex 0 if male, 1 if female	-4	0.0183	<1	⬇ RISK
strain Vero, Palm	0.01	1.01	~1	— RISK

HR = 1.8 for 1% vs. 0.5% permethrin
→ 80% higher risk

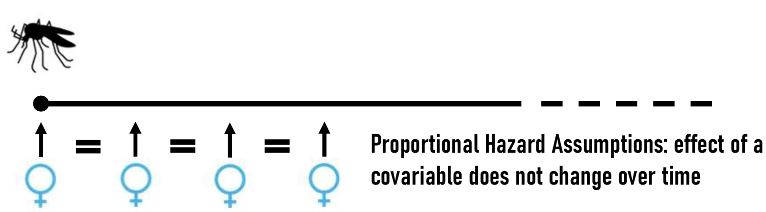


21

Cox Proportional Hazard Model Assumptions

Assumptions

- **Non-informative censoring**
- **Proportional hazards**
- **Linearity** of continuous covariates
  i.e., effect of a covariate is constant over time



22

Survival Analysis – Final Considerations

- Censored data are common in ecological studies – make sure they're accounted for.
- Kaplan-Meier curves are visual, but log-rank and Cox regression provide statistical rigor.
- Check assumptions! Choose other methods for time-to-event analysis if necessary, e.g.
  - Restricted mean survival time (RMST)
  - Accelerated Failure Time (AFT) regression models
  - Parametric survival models
    (see e.g., overview of Parametric Survival Modeling)

23